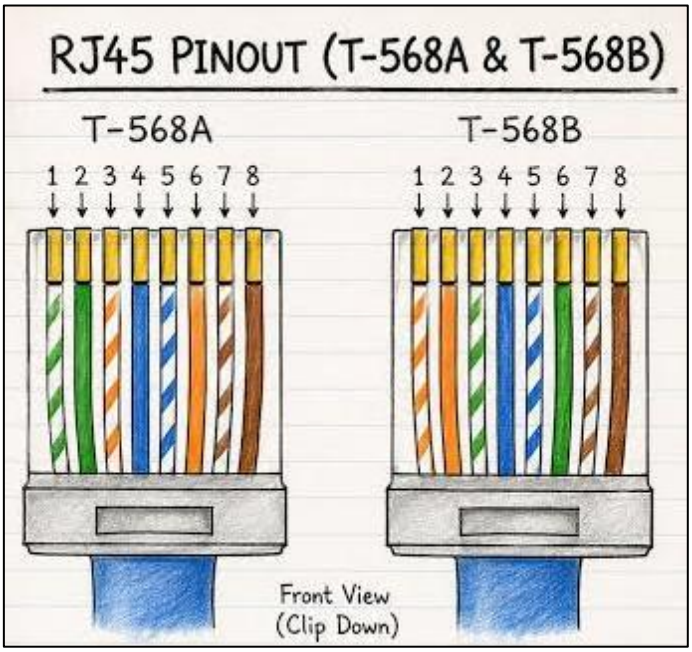
1. Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards.



Pin No.	T568A Color Code	T568B Color Code
Pin 1	White/Green	White / Orange
Pin 2	Green	Orange
Pin 3	White / Orange	White/Green
Pin 4	Blue	Blue
Pin 5	White / Blue	White / Blue
Pin 6	Orange	Green
Pin7	White / Brown	White / Brown
Pin 8	Brown	Brown

2. Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.

Feature	**10BASE-T (Old Ethernet)**	**100BASE-TX (Fast Ethernet)**	**1000BASE-T (Gigabit Ethernet)**
Speed	Slow (10 Mbps)	Medium (100 Mbps)	Fast (1000 Mbps or 1 Gbps)
Cable Type	Cat3 (Very old phone lines)	Cat5 (Older internet cable)	Cat5e or Cat6 (Modern internet cable)
Max Length	100 metres	100 metres	100 metres
Plug type	Standard RJ45 plastic clip	Standard RJ45 plastic clip	Standard RJ45 plastic clip
How it works	Uses 4 wires inside	Uses 4 wires inside	Uses all 8 wires inside to boost speed

3. Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why.

Hint: Think about how transmit and receive pairs are wired in each cable type.

To connect two laptops directly, the required cable depends on how old the the laptops are.

Today, a regular straight-through cable can be used as well.

(A) *Problem: Why they need to cross over.*

Inside every laptop's internet port, specific wires are used to **send** data (the "mouth") and other wires are used to **receive** data (the "ear").

- If a regular straight-through cable is used , the mouth of the first laptop connects straight to the mouth of the second laptop
- Since both laptops are trying to talk into each other's mouths and listen to each other's ears, they cannot communicate.

(B) *Solution:*

- An older setup needs a crossover cable. This cable physically crosses the wires inside. It connects the mouth (transmit) of the first laptop to the ear (receive) of the second laptop so they can talk and listen properly.

(C) *Why Straight-Through Works Today*

Today we can just use a regular straight-through cable.

Modern laptops have a smart feature called Auto-MDIX. When you plug in a regular cable, the laptops realize they are connected directly to each other. The laptops then electronically flip their own internal wires. They change their mouths to ears and ears to mouths automatically so the data flows perfectly.

4. List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.

(A) Smart TV

- Likely Standard: 100BASE-TX (Fast Ethernet)
- Simple Explanation**:** This standard handles speeds up to **100 Mbps**. Because streaming high-quality 4K movies only requires about 25 to 50 Mbps, TV companies save money by using this older, slower network speed. It is plenty of speed for watching videos without buffering.

(B) Gaming PC or Desktop Computer

- Likely Standard: 1000BASE-T **(Gigabit Ethernet)**
- Simple Explanation**:** This standard handles speeds up to **1,000 Mbps** (1 Gbps). Computers need much more data than TVs for downloading large files, playing online games without lag, and sharing files quickly with other computers.

(C) College Wi-Fi Router (Access Point)

- Likely Standard: 2.5GBASE-T (Multi-Gigabit Ethernet)
- Simple Explanation: This standard handles speeds up to 2,500 Mbps (2.5 Gbps). Because a college Wi-Fi box has to provide internet to dozens of students at the exact same time, it needs a much bigger "pipe" so that everyone's phone and laptop stays fast.